

Problem 1

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Problem 1 Differentiate the functions using product or quotient rule.

(a) $h(u) = 7ue^u$

Explanation. Applying the power rule:

$$\begin{aligned} h(u) &= 7ue^u \\ \implies h'(u) &= (7ue^u)' \\ &= 7 \cdot e^u + 7u \cdot e^u \end{aligned}$$

(b) $s(t) = \frac{\sqrt{t}}{e^t}$

Explanation. Applying the quotient rule:

$$\begin{aligned} s(t) &= \frac{\sqrt{t}}{e^t} \\ \implies s'(t) &= \left(\frac{\sqrt{t}}{e^t} \right)' \\ &= \frac{\frac{1}{2}t^{-1/2} \cdot e^t - \sqrt{t} \cdot e^t}{e^{2t}} \end{aligned}$$